Mario Sinclair

Tampa, FL | mario.sinclairbz@gmail.com | linkedin.com/in/mario-sinclair | github.com/MarioSinclair

Education

University of South Florida GPA: **3.81**/4.0

Bachelor of Science in **Computer Science** Dean's List: Fall 2023, Spring 2024, Fall 2024 May 2027

Coursework: Data Structures, Algorithms, Software Engineering, Generative AI, Computer Vision, Control of Mobile Robots

Experience

Delta Air Lines, Inc May 2026 – Aug 2026

Data Engineering Intern, Compliance System Management *Atlanta, GA*

- Built and maintained **SQL** ELT pipelines in **Palantir Foundry** for maintenance data across **900+ aircraft**, feeding the reports and dashboards engineering teams use every week.
- Wrote a **TypeScript** tool that automated the team's manual aircraft-bridging process, saving about **15 hours a week**.
- Wrote a **Python** validation framework to catch bad records, and sped up **Power BI** dashboard queries by **30%**.
- Moved three dashboards to a new source table, rewriting the **SQL** and checking every number against past results.
- Built **Python** tools with **Kiro**, like a tracker that matches parent and child engine records.

UG Research Assistant - Reality, Autonomy and Robot Experience (RARE) Lab Sep 2025 - Present

Python, C++, ROS1/ROS2, Nav2, Gazebo, PyTorch, Docker, Linux *Tampa, FL*

- Working on projected AR with **CompenHR**, a deep learning method that corrects projected images on uneven surfaces, and mounting it on a Fetch, a Unitree Go2, and a Unitree R1.
- Built a **ROS2/Nav2** navigation stack for a Unitree Go2 robot dog, writing the **Python** launch files and **C++** code that sends goals and tracks progress. After tuning, it reached its goal in **7 of 10** real-world trials.
- Wrote a **C++** action server and client that moves a Fetch robot a set distance using odometry.

IEEE Robotics & Automation Society, NASA Lunabotics Team Aug 2026 - Present

Software Team, Autonomous Navigation *Tampa, FL*

- Leading the software architecture for an autonomous lunar excavation robot, designing the **ROS 2** system that links **Nav2**, **FAST-LIO2** LiDAR SLAM, and **micro-ROS** motor control on a Teensy 4.1.

Projects

Sherpa, Safe Campus Navigation (HackGT 2026) Sep 2026

Python, Flask, NetworkX, GeoPandas, JavaScript, MapLibre GL, PostgreSQL (Supabase), Render

- Built a night-walking navigation app for Georgia Tech on a 2-person team, writing all of the software (live at **appsherpa.tech**).
- Wrote the routing engine using weighted **Dijkstra** over a **10.6k-edge** campus map with **3,600+ streetlights** and accessibility data. Safest routes had **63%** less poorly lit walking for about **8.5%** more distance.
- Found and fixed errors in the campus map data, which raised stair-free routes from **67% to 80%** across **351 routes**.

TriageBot, LLM Agent for Bug Triage Oct 2026

Python, Claude API (tool use), GitHub API

- Built an **LLM agent** on the **Claude API** that sorts my lab's GitHub issues by category and severity and flags duplicates. Tested on **300 closed issues**: **78%** top-1 accuracy and **97%** duplicate detection.

PPE Compliance Checkpoint, Real-Time Computer Vision Sep 2026

Python, PyTorch, YOLO11, OpenCV, dlib, FastAPI, SQLite

- Trained a **YOLO11** model on **1,200+** hand-labeled boxes to check 4 pieces of safety gear at about **20 FPS** on a laptop CPU, with face recognition to identify each worker before a gate opens.

OptionScope, Options Pricing Tool May 2025 - Present

Python, PySpark, Databricks, Apache Airflow, SQL, FastAPI, React

- Built a **Black-Scholes** pricing tool that charts the Greeks, fed by a **PySpark** job scheduled with **Airflow**.

EquiBull AI, 2nd Place at InnoTech AI Hackathon (NSBE 2025) Oct 2025

Python, R, Vertex AI, Power BI, Tableau

- Analyzed **1,000+ HMDA mortgage records** and fine-tuned Gemini 2.5 Flash to predict credit risk for a credit readiness platform.

Leadership

Google Developer Group | Finance Chair Jun 2026 - Present

- Secured **$9K** in sponsorships for HackUSF 2026 and DevFest 2025, and manage the chapter's budget.

Society of Hispanic Professional Engineers | Technical Director Sep 2025 - Apr 2026

- Built the websites for USF's largest hackathons (**410+**) and taught React to **70+** students.

National Society of Black Engineers | Conference Planning Chair Jan 2025 - Apr 2026

- Led convention planning for **30+ students** and organized Engineering Expo outreach for **5,000+ K-12 students**.

Skills

Languages: Python, C++, C, TypeScript, JavaScript, SQL, R, Bash **Spoken:** English, Japanese

Data: Palantir Foundry, PySpark, Databricks, Apache Airflow, PostgreSQL, MySQL, Power BI, Tableau, pandas

Web & AI: React, Node.js, Flask, FastAPI, REST APIs, PyTorch, YOLO, OpenCV, Claude API, Claude Code, Kiro

Robotics & Tools: ROS1/ROS2, Nav2, Gazebo, Jetson, ESP32, Docker, Linux, Git, GitHub Actions, AWS